ML in Financial Compliance

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Outline

- Financial Compliance: AML/ CBT
- Unsupervised methods
  - Clustering
  - Graphical ML

Aspects of Financial Compliance

- *The regulation and enforcement of the laws and rules in finance and the capital markets for companies.*

- Two main aspects
  - **Regulatory** compliance
    - Regulatory for eg Basle Accord capital calculations, international sanctions
  - **Financial Crime** compliance
    - Guard against AML, CBT and fraud detection.

Battle against Crime and Making the World a Safer Place

- Protecting the global financial system https://www.ft.com/content/a81bc993-97fc-45d2-bd55-9dd32b5957d

Tightens scrutiny of clients

Singapore banks tighten scrutiny of clients after money-laundering scandal

Customers from China in spotlight as waiting times stretch to four months to open accounts

The Monetary Authority of Singapore has instructed financial institutions to scrutinise people and companies linked to 10 suspects in a S2.4bn money laundering probe © Roslan Rahman/AFP/Getty Images

Mercedes Ruehl in Singapore

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https://www.ft.com/content/8d1a9820-05ac-4f90-9417-8efd2222c42c

Financial Crime Compliance

- Anti-Money laundering (CBT)
  - Laundering of ‘dirty money’ in the world financial systems
  - Source of funds from illicit activities – drugs, slavery, scams etc.

- Combat Terrorism financing (CBT)
- Scams / Frauds alert!

Anti-Money Laundering (AML) & Combating the Financing of Terrorism (CFT)

- Protecting the global financial system

The process of making illegally obtained money appear legitimate

1. Placement

Introduction of illegal funds into the financial system through deposits, wire transfers, or other means

2. Layering

Complex transactions to obscure the origin of funds through multiple transfers, conversions, or investments

3. Integration

Reintroduction of laundered funds into the legitimate economy as apparently legal wealth

Why AML/CBT matters: Global Impact

Economic Impact

- Undermines financial system integrity
- Distorts investment and trade flows
- Estimated 2-5% of global GDP laundered annually

Social Impact

- Funds organized crime and terrorism
- Increases corruption and inequality
- Threatens national security

Regulatory Impact

- Reputational damage to institutions
- Severe penalties and sanctions
- Loss of banking licenses

Global Cooperation

- Cross-border coordination essential
- Information sharing critical
- Standardized regulations needed

Key International Framework: FATF and global standards

Financial Action Task Force (FATF)

Established in 1989, FATF sets international standards for combating money laundering and terrorist financing

FATF 40 Recommendations Cover:

Risk assessment and policy coordination

Customer due diligence (KYC)

Suspicious transaction reporting

International cooperation

Beneficial ownership transparency

Targeted financial sanctions

Core AML/CFT Obligations
What financial institutions must do

Know Your Customer (KYC)
Verify customer identity, understand business relationships, and assess risk profiles
Customer Due Diligence (CDD)
Ongoing monitoring of transactions and periodic review of customer information
Suspicious Activity Reporting (SAR)
Report unusual or potentially criminal transactions to relevant authorities
Record Keeping
Maintain comprehensive records of transactions and customer data for specified periods
Risk-Based Approach
Apply enhanced measures for higher-risk customers, products, and jurisdictions

Red Flags & Warning Signs
Common indicators of suspicious activity

Transaction Patterns
- Unusual transaction volumes or frequencies
- Transactions just below reporting thresholds
- Rapid movement of funds between accounts
- Complex transactions with no clear purpose
- Use of multiple accounts or intermediaries

Customer Behavior
- Reluctance to provide information
- Inconsistent business rationale
- Unusual concern about reporting requirements
- Transactions inconsistent with profile
- Use of shell companies or nominees

Red Flags & Warning Signs
Common indicators of suspicious activity

Geographic Indicators
- Transactions involving high-risk jurisdictions
- Frequent cross-border wire transfers
- No apparent business connection to location
- Sanctions-listed countries involvement

Product/Service Misuse
- Cash-intensive business anomalies
- Unusual use of trade finance instruments
- Cryptocurrency mixing or layering
- Prepaid cards with large balances

How AI/ML helps in financial compliance

- Sifting through the deluge of data to find patterns and anomalies.
- Eg:
  - Transactions monitoring: distinguish genuine suspicious activity (fraud detection) from routine transactions more accurately. Reducing false positives top priority. Huge false positives create massive backlogs.
  - Blockchain analytics: Tools to trace cryptocurrency transactions and identify connections to illicit activities
  - Customer risk assessment: during onboarding and KYC (know your customer); helps to uncover hidden links
  - Automated (continued) risk assessment: constant monitoring of news and regulatory updates through NLP (for e.g. enable summarization) on key entities

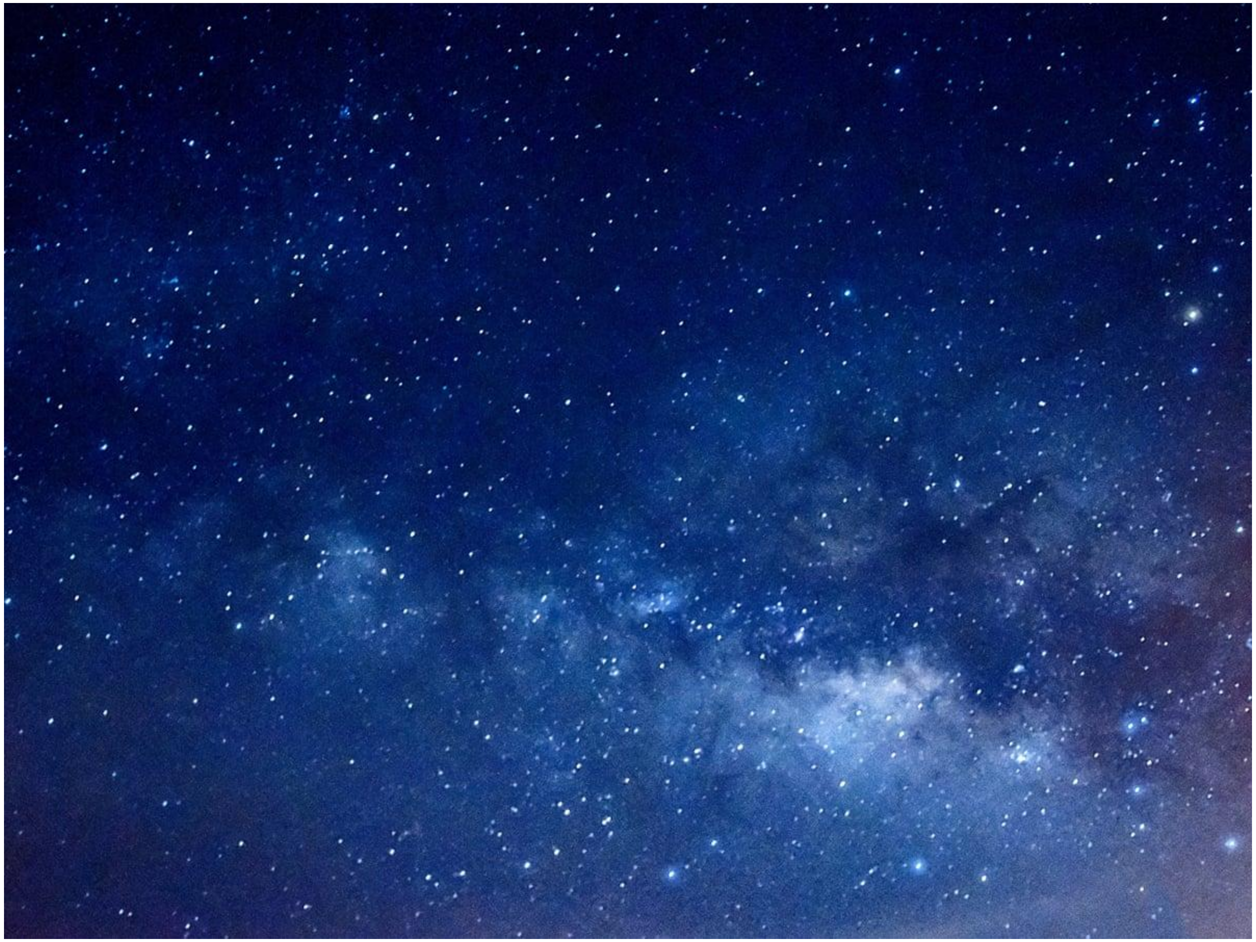
AI/ML Techniques for Financial Compliance

- Unsupervised ML methods:
  - K-means clustering
  - DBSCAN
  - Graphical ML

- Objectives of unsupervised learning usually to reduce the complexity of the data/ reduce noise. Done by: (usu.)
  - ***Transform*** data points
  - ***Reduce*** the no of data points

- How does it capture key essence of original data?
  -> Through grouping (clustering) of data points

Unsupervised Learning



Look up to sky and observe the stars

What do you see?

Constellations!

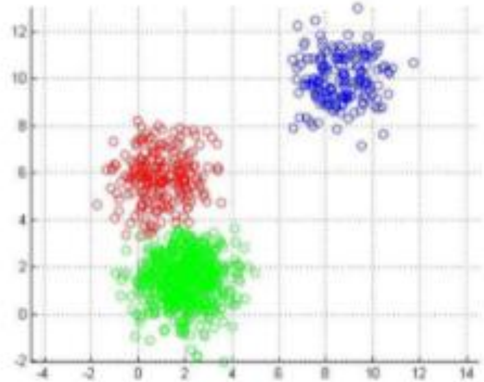
Stars that are grouped together – **constellations**!

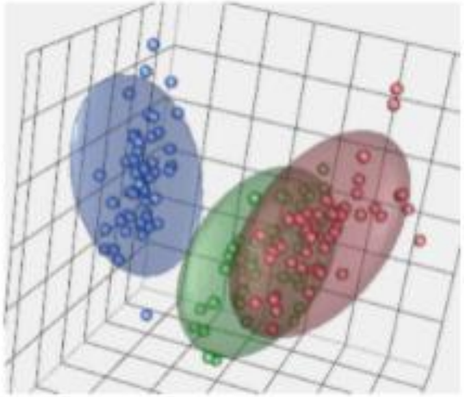
K-means clustering

- How do you group multiple ‘entities’ together based on their characteristics? *Fixed no of groups* at start.

- Eg. Customers with features – age, income, occupation, interests etc

- Objectives:
  - group members as similar as possible to one another
  - groups as different as possible with other groups





How to measure Similarity?

Similarity (Distance) Measure

- A distance measure is normally used to measure the similarity or dissimilarity between two data objects
- Data is often in matrix
  - *n* records (rows), each with *p* attributes/ features in columns
  - *n*-by-*p* matrix structure
  - x_{ab} = value for a^{th} record and b^{th} attribute

Attributes

Records

$$
\begin{array}{r}
record\ 1 \\
\\
record\ i \\
\\
record\ n
\end{array}
\begin{bmatrix}
x_{11} & \cdots & x_{1f} & \cdots & x_{1p} \\
\cdots & \cdots & \cdots & \cdots & \cdots \\
x_{i1} & \cdots & x_{if} & \cdots & x_{ip} \\
\cdots & \cdots & \cdots & \cdots & \cdots \\
x_{n1} & \cdots & x_{nf} & \cdots & x_{np}
\end{bmatrix}
$$

Similarity (Distance) Measure

- General distance measuring method: *Minkowski* distance

$$d(i,j) = \sqrt[q]{(|x_{i_1} - x_{j_1}|^q + |x_{i_2} - x_{j_2}|^q + \ldots + |x_{i_p} - x_{j_p}|^q)}$$

where $i = (x_{i1}, x_{i2}, \ldots, x_{ip})$ and $j = (x_{j1}, x_{j2}, \ldots, x_{jp})$ are two *p*-dimensional data objects, and *q* is a positive integer

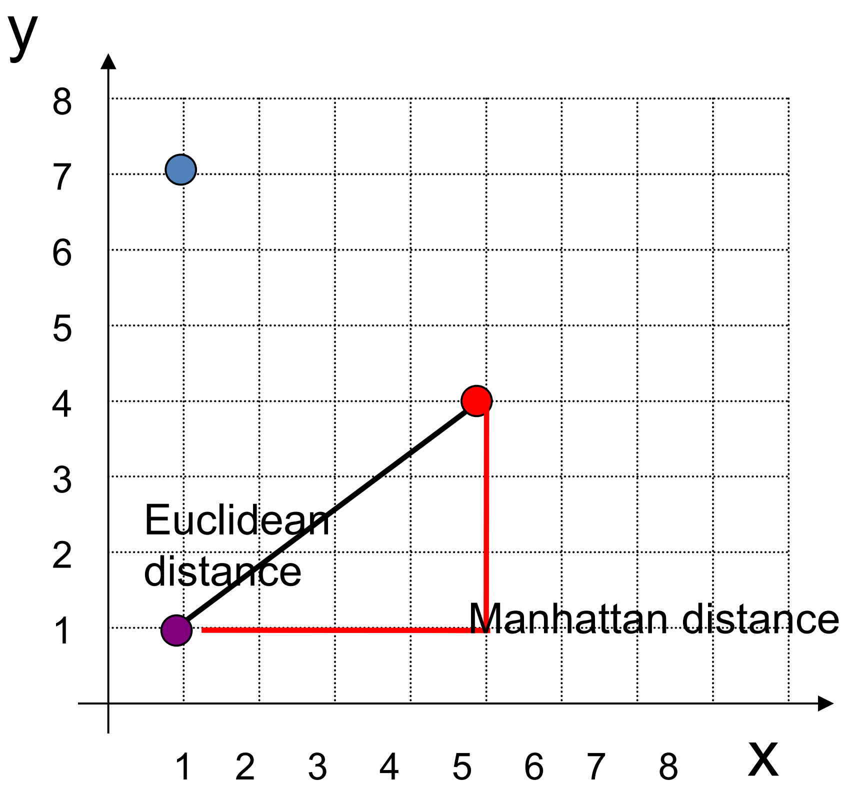
- For Euclidean distance: q = 2

$$d(i,j) = \sqrt{(|x_{i_1} - x_{j_1}|^2 + |x_{i_2} - x_{j_2}|^2 + \ldots + |x_{i_p} - x_{j_p}|^2)}$$

- Manhattan distance: q = 1

$$d(i,j) = |x_{i_1} - x_{j_1}| + |x_{i_2} - x_{j_2}| + \ldots + |x_{i_p} - x_{j_p}|$$

Euclidean & Manhattan



Distance between red and purple?

Transform Numeric Data

Step 1: Standardize the data

- To ensure they all have equal weight
- To match up different scales into a uniform, single scale

Step 2: Compute dissimilarity between records

Euclidean, Manhattan or Minkowski distance

Nominal (Categorical) Data

- Distance between Nominal variable (more than 2 states e.g., red, yellow, blue, or green) are computed by:
  - Method 1: Simple matching

  m: # of matches, p: total # of variables

$$d(i,j) = \frac{p-m}{p}$$

  - Method 2: use a large number of binary variables

  creating a new binary variable for each of the M nominal states

Popular ML Term: Anomaly Detection

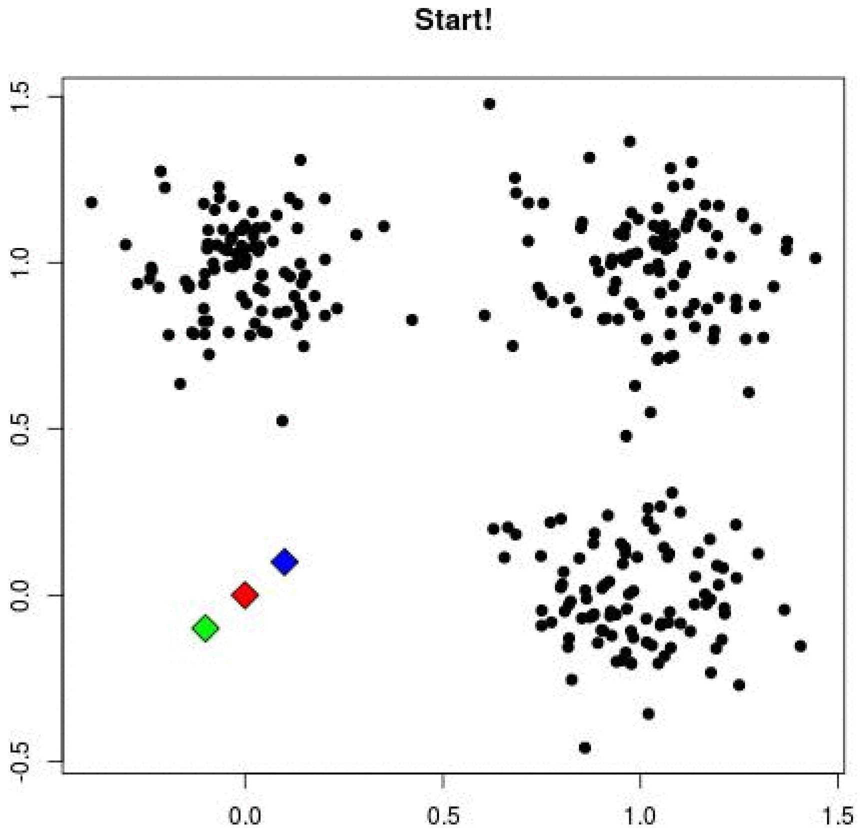
- Anomaly: also called outlier detection.
    - what’s not ‘belong’ to usual groups or abnormal.
    - Commonly uses unsupervised learning.

- Used in finance for fraud detection, suspicious trades, AML, trading, stress testing, market and trading anomalies

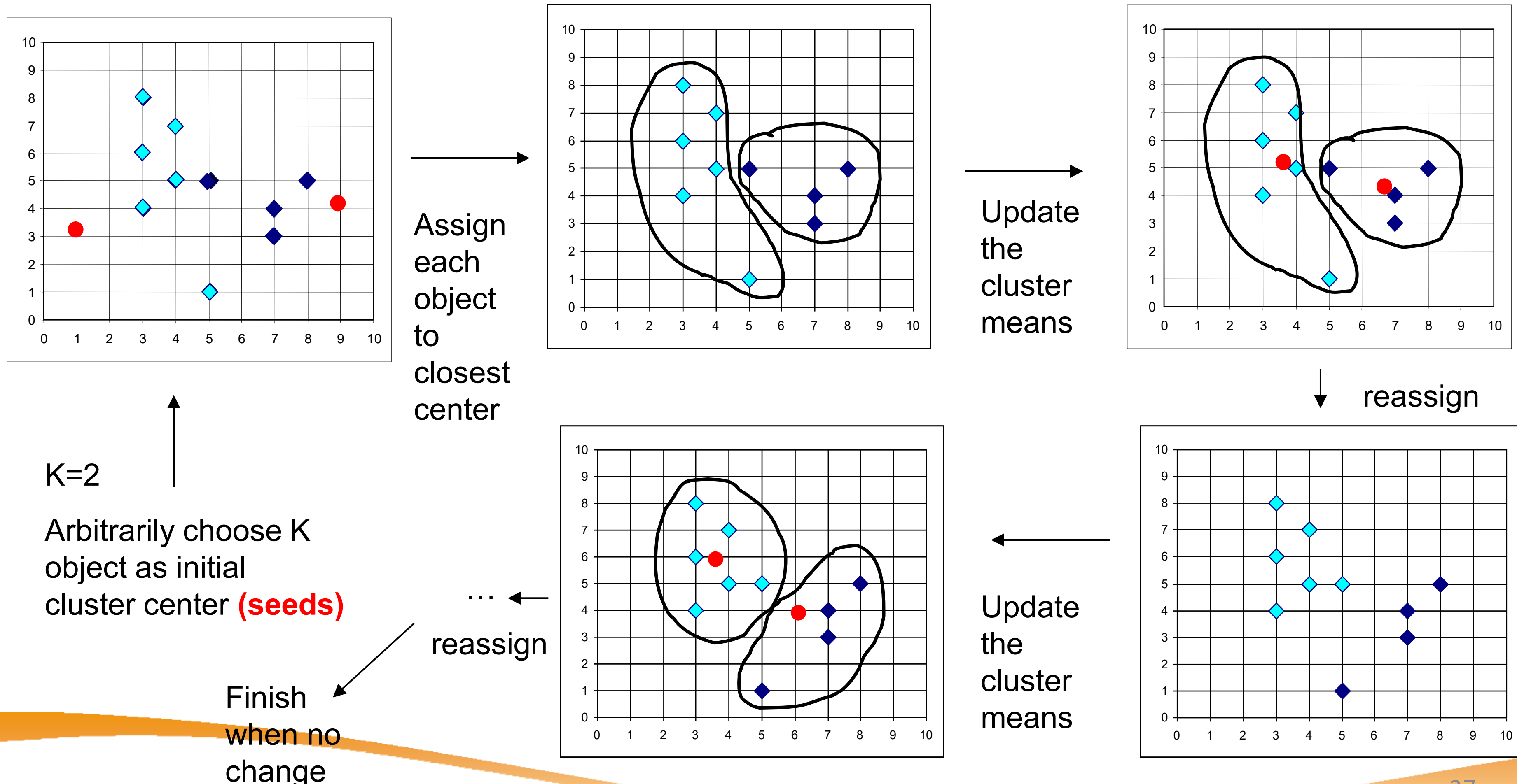
k-Means Algorithm

- Given *k*, the *k-means* clustering works as follows:
  1. Choose a value for *k*, the total number of clusters.
  2. Randomly choose *k* points as cluster centers (seeds)
  3. Assign the remaining instances to their closest cluster center.
  4. Calculate a new cluster center for each cluster.
  5. Repeat steps 3-4 until the cluster centers do not change.

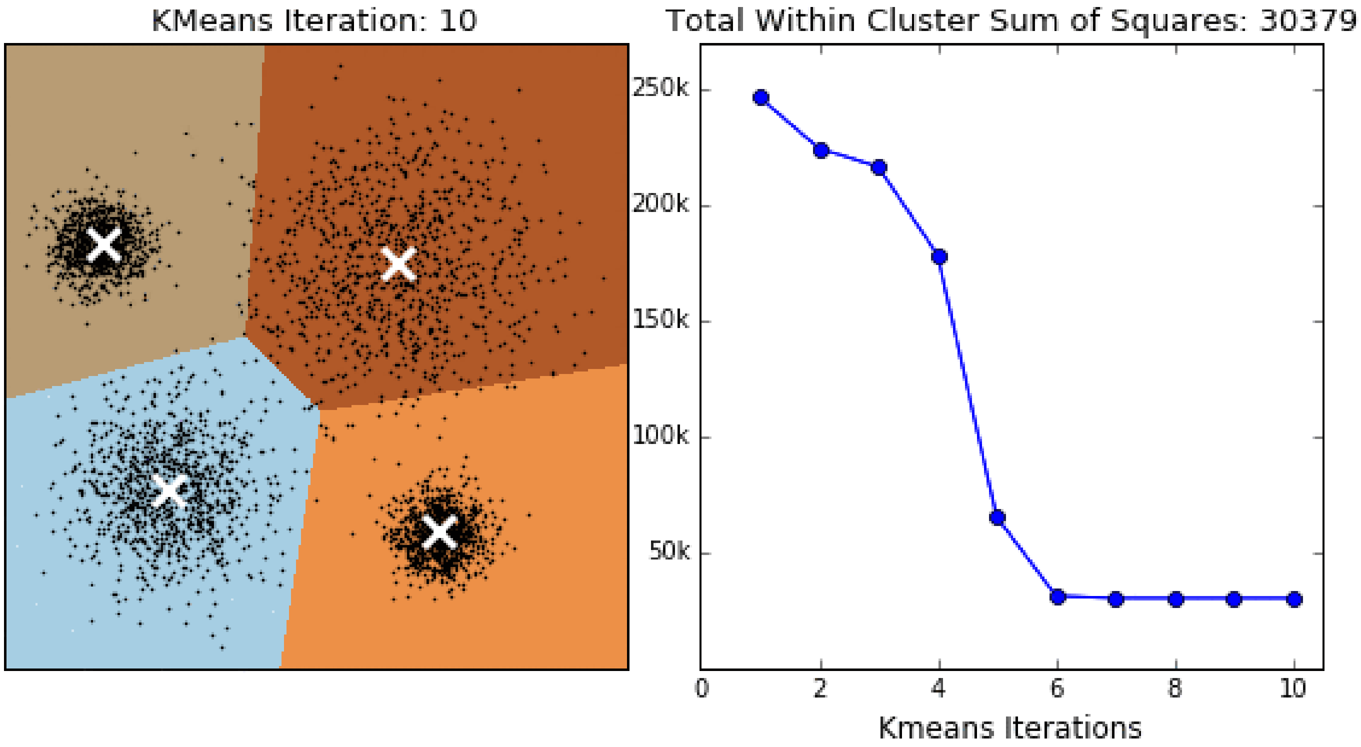
k-Means Clustering in action



k-Means Clustering



k-Means Clustering

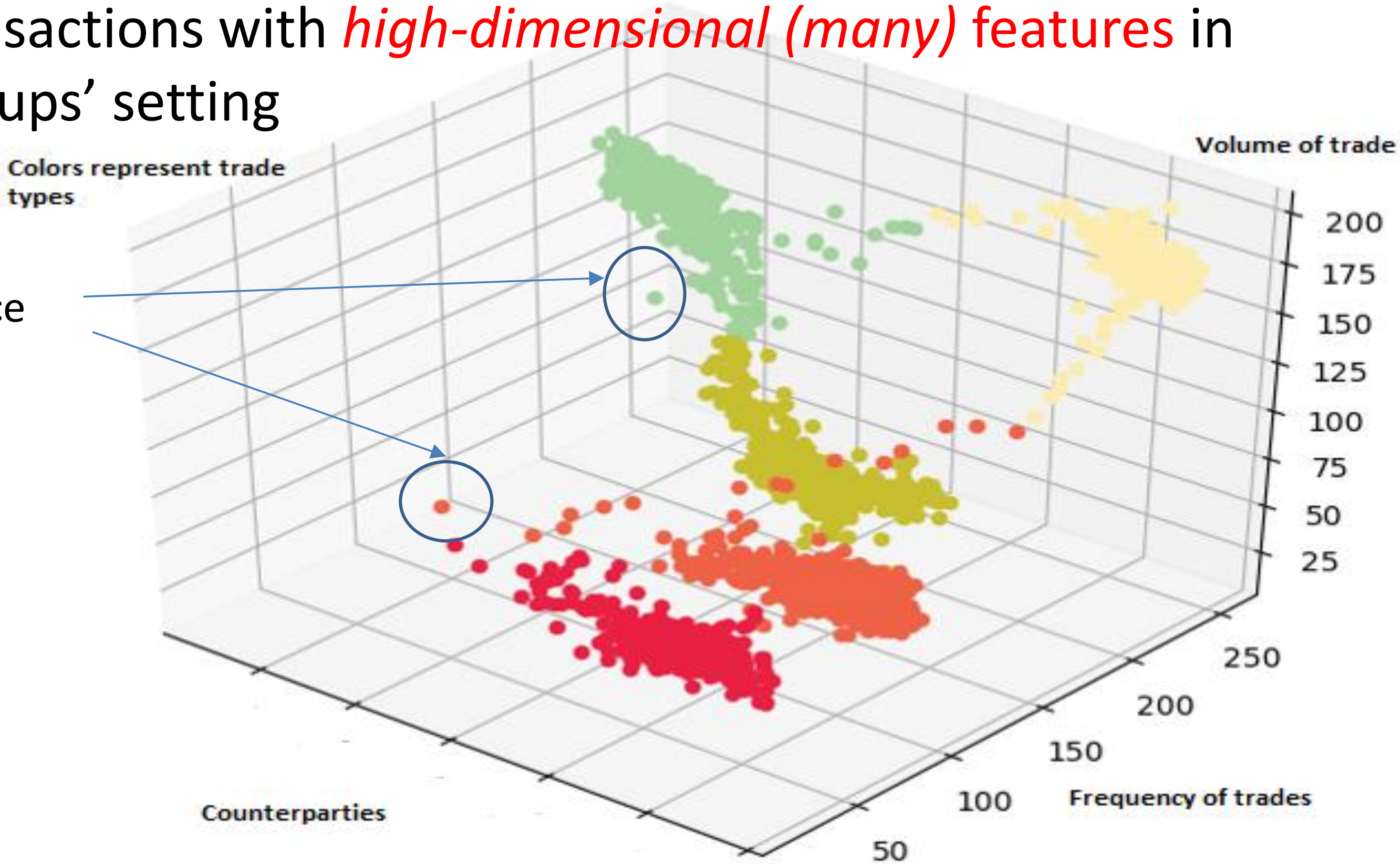


With more iterations, the data points within each cluster become 'more similar' as a whole

K-means clustering in fraud detection

- Anomaly detection for money-laundering, frauds (in credit card use & applications etc)
  - Early days use rules-based ML.
  - Idea is to find *non-conforming clichés* amongst trades/ transactions with *high-dimensional (many) features* in 'groups' setting

Non-conformance



Density-based Clustering

- **Clustering based on density** (local cluster criterion), such as the density of points
  - Earlier, with K-means/ medoids: distances are measured wrt from points to *a point*; but now to a *group of nearest points*

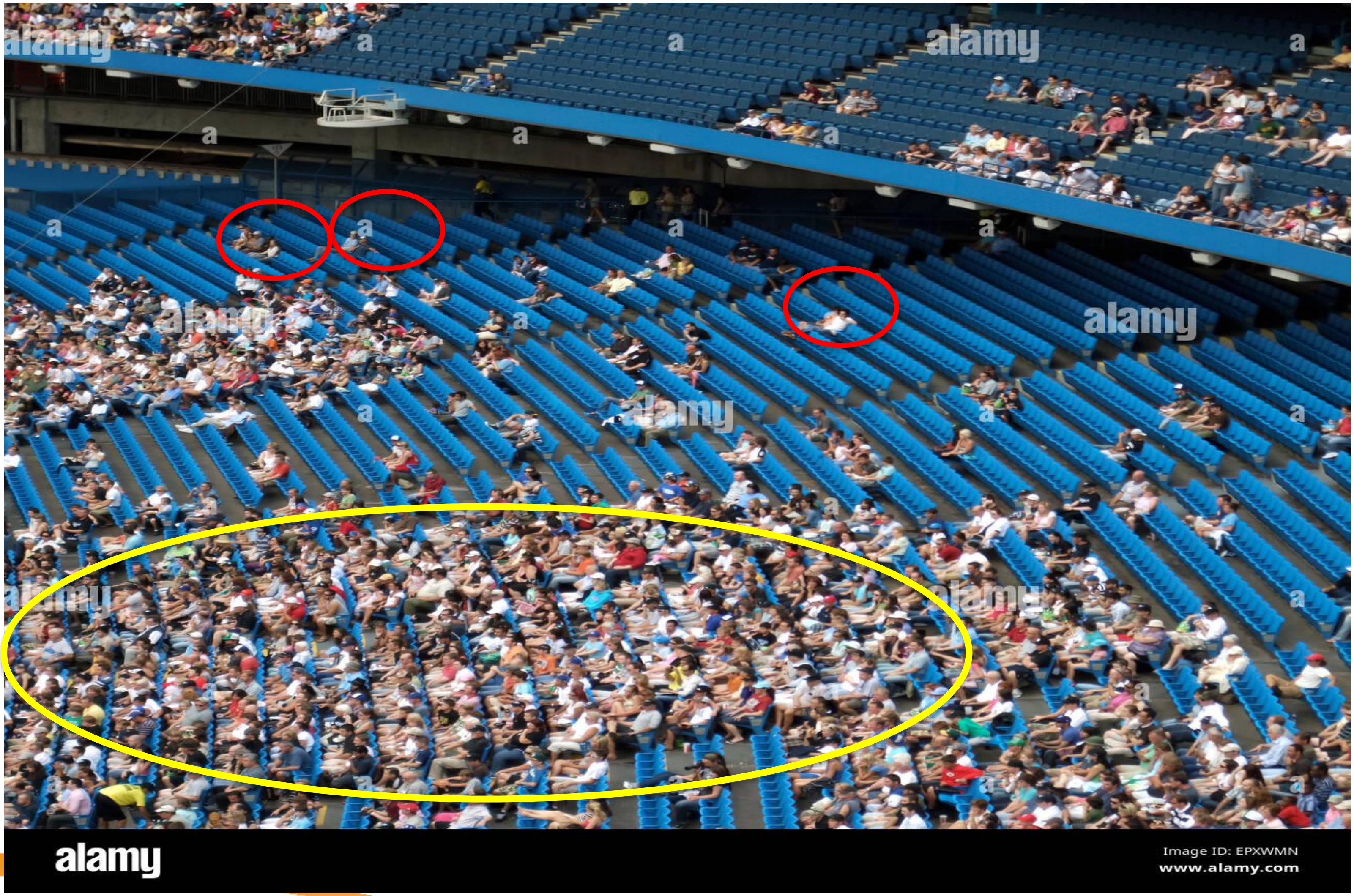
Major features:

Able to uncover clusters of arbitrary shape

Handle noise

Need density parameters as termination condition

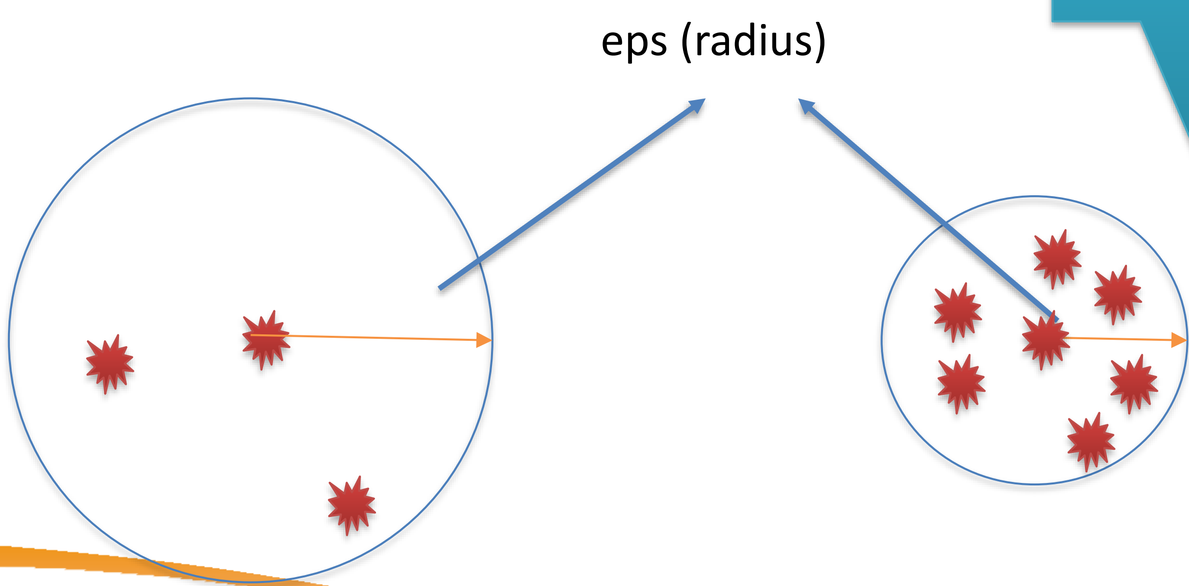
DBScan – an analogy in the stadium



How is ‘density’ defined?

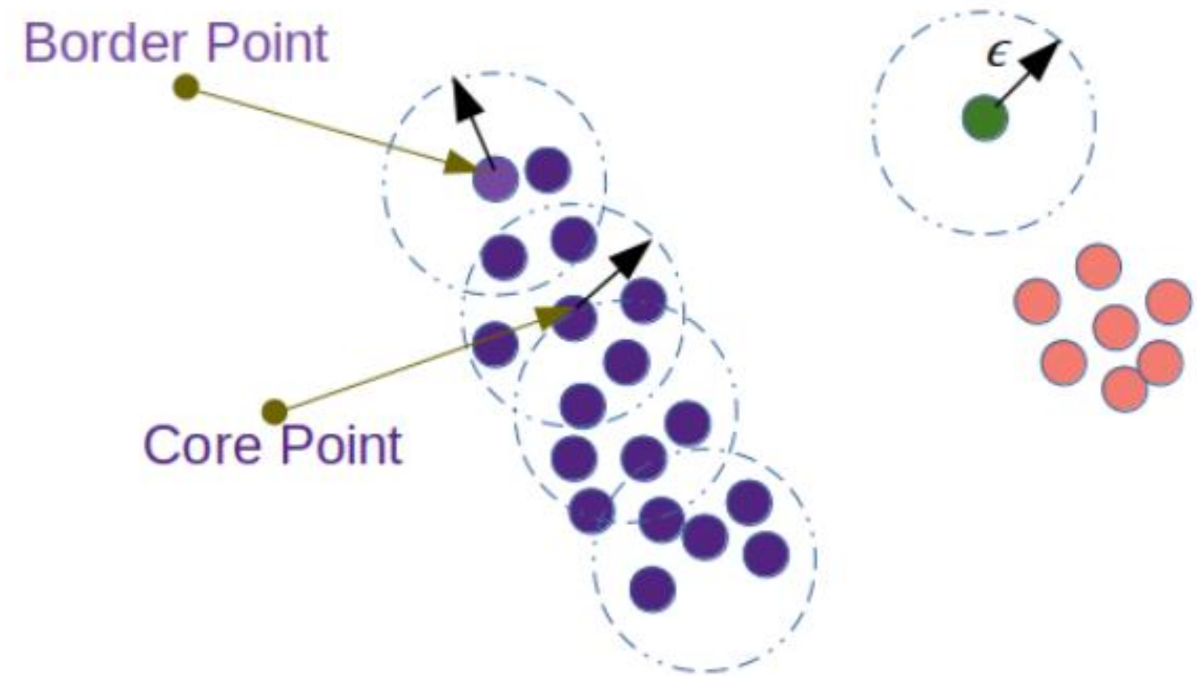
- Two main *hyper-parameters* to define density
  - Eps (radius)
  - minPts (minimum number of points in a group)

Dense enough? ->
Regard as a group



DBScan – Group or Noise

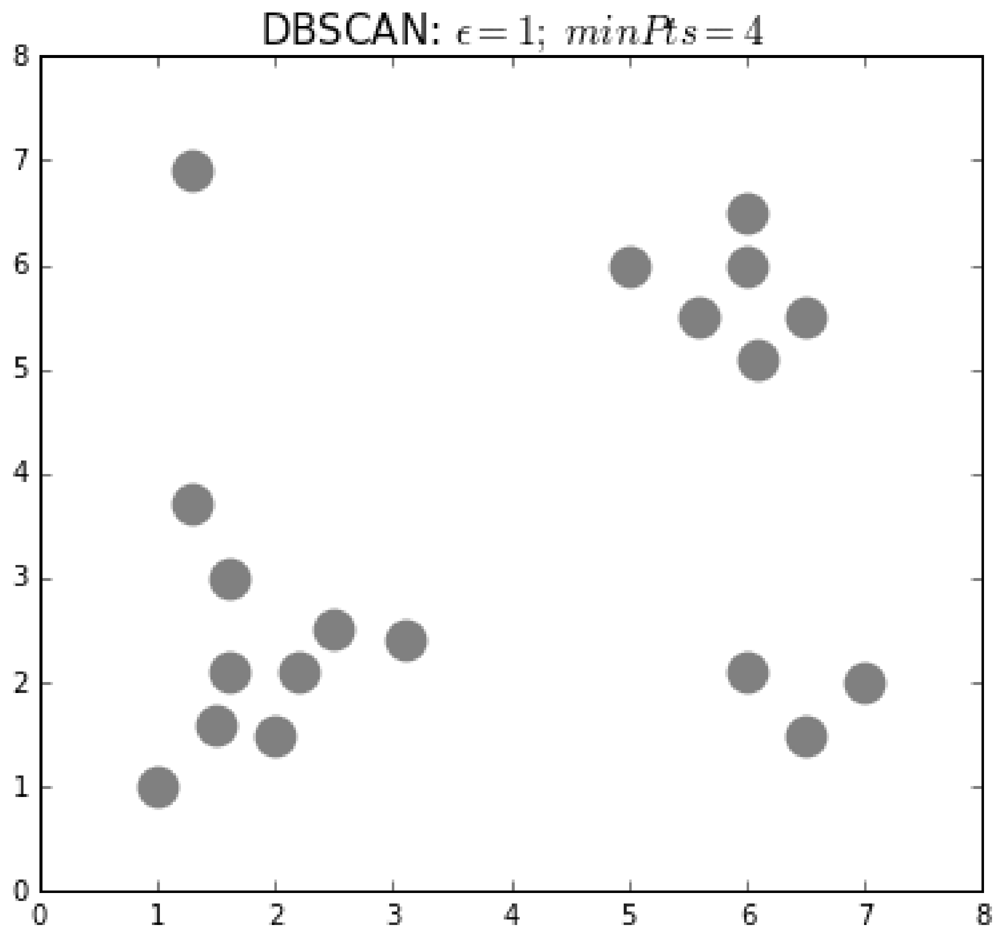
- Group data points as either:
  - Part of a group
  - Noise (outside group)



Use of DBSCAN

- Fraudulent trades
- In the past, rules-based method to identify fraudulent trades AML/CBT.
- Normally works by looking single characteristics
  - Trade notionals
  - Trade frequencies etc
- DBSCAN/ other ML Methods useful in that it considers high-dimensional features, and in bias-free method.

DBSCAN Demonstration



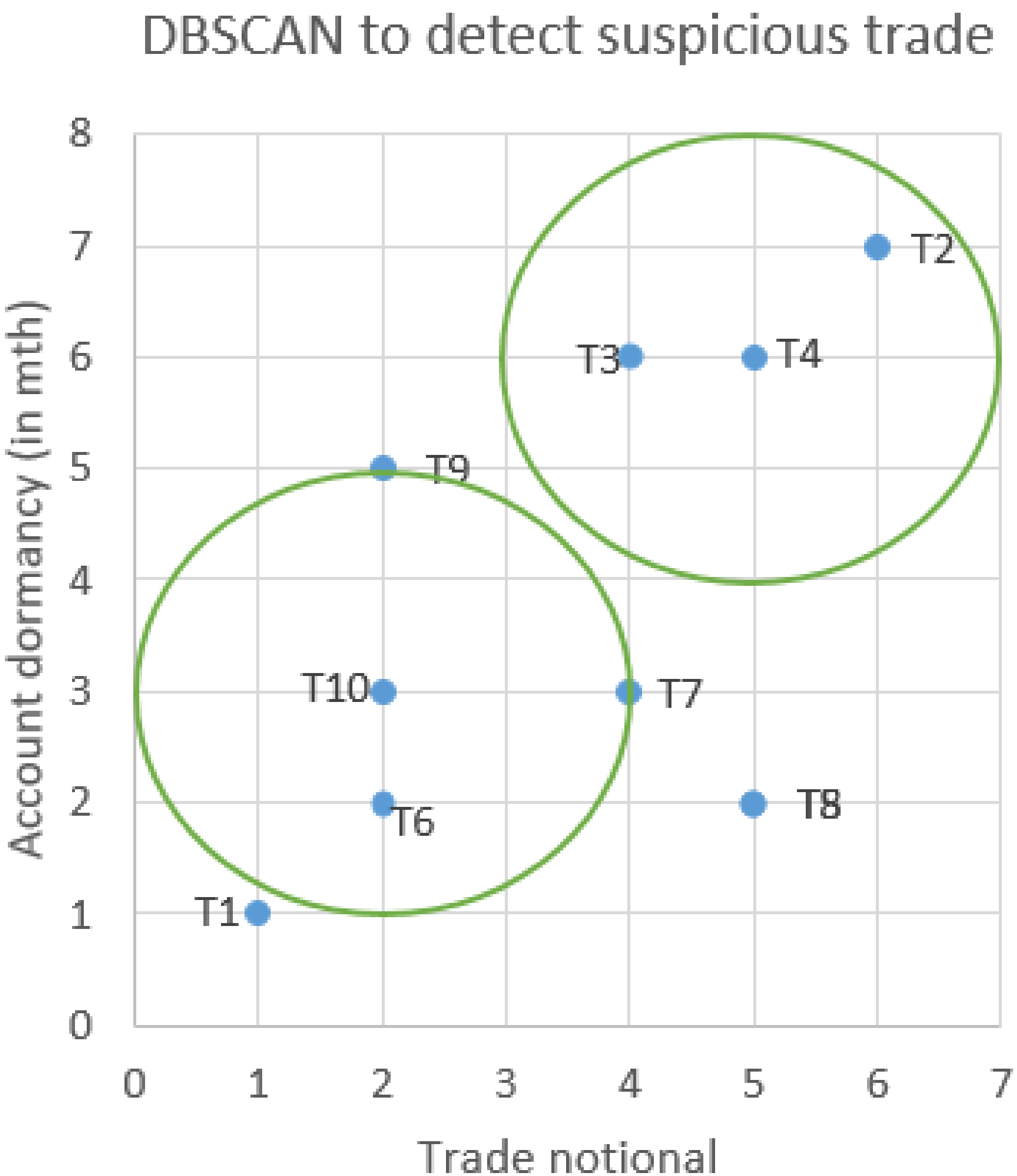
https://dashee87.github.io/data%20science/general/Clustering-with-Scikit-with-GIFs/

Advantages of using ML for compliance

- Rules are static but ML is dynamic.

- Tackle high-dimensional transaction features at once
- ML learns by ‘itself’. Rules etc. need *historical precedence* and human judgement

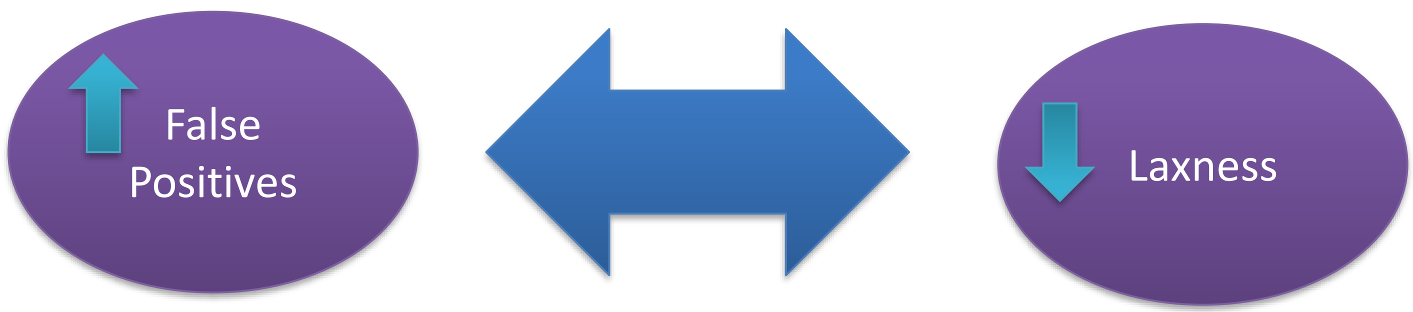
Test your understanding: menti.com xxxxxx



For illustration, we use only 2 features to identify suspicious trades:
For eps = 2 and minPts = 4; what are the trades flagged out as suspicious?

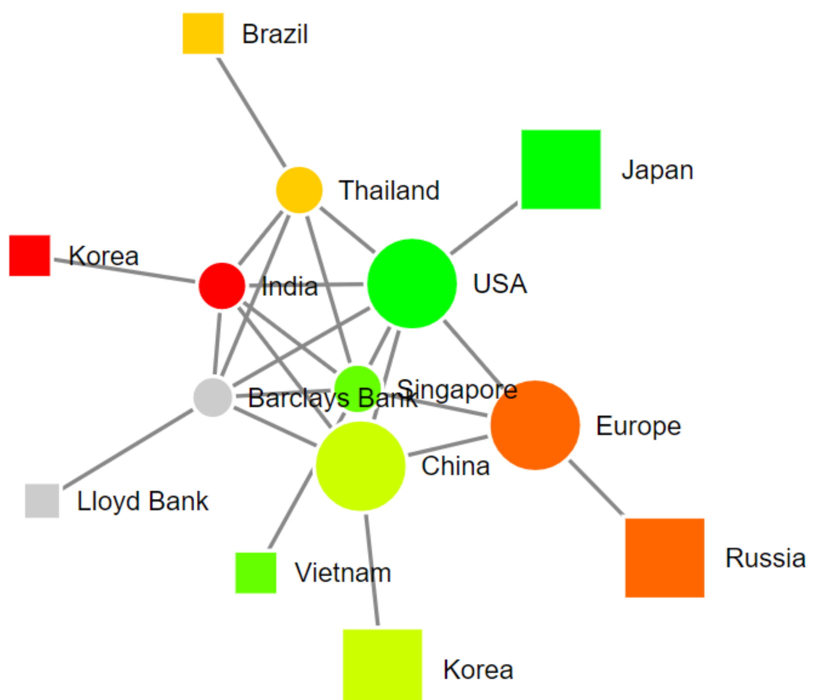
False Positives (Suspicious) Trades

- Previous eg of: too many trades flagged out?
- Raise (stricter) or lower the bar? (ML Metrics) -> tug of war between regulators and FIs



Introducing and Demystifying Graphs

- Put simply, graph are a network of *nodes* and *links*. Links connect the nodes.
- The links represent relationships. Relationships are ubiquitous and key in the financial industry.



Graphs – more common than thought

Common in finance. Examples:

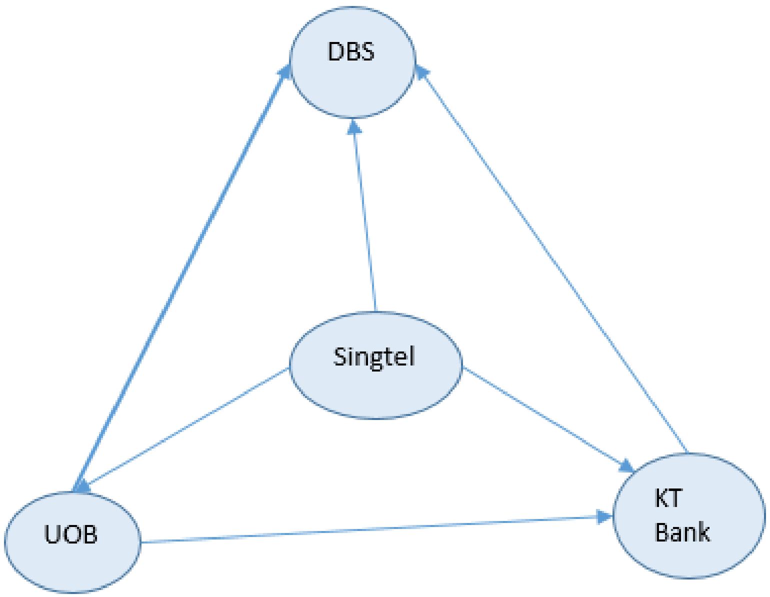
- Trade links, debt inflow/ outflow
- Correlation matrices in portfolio risk management.

More in the applications on finance … …

- A typical correlation matrix can be represented as a graph.

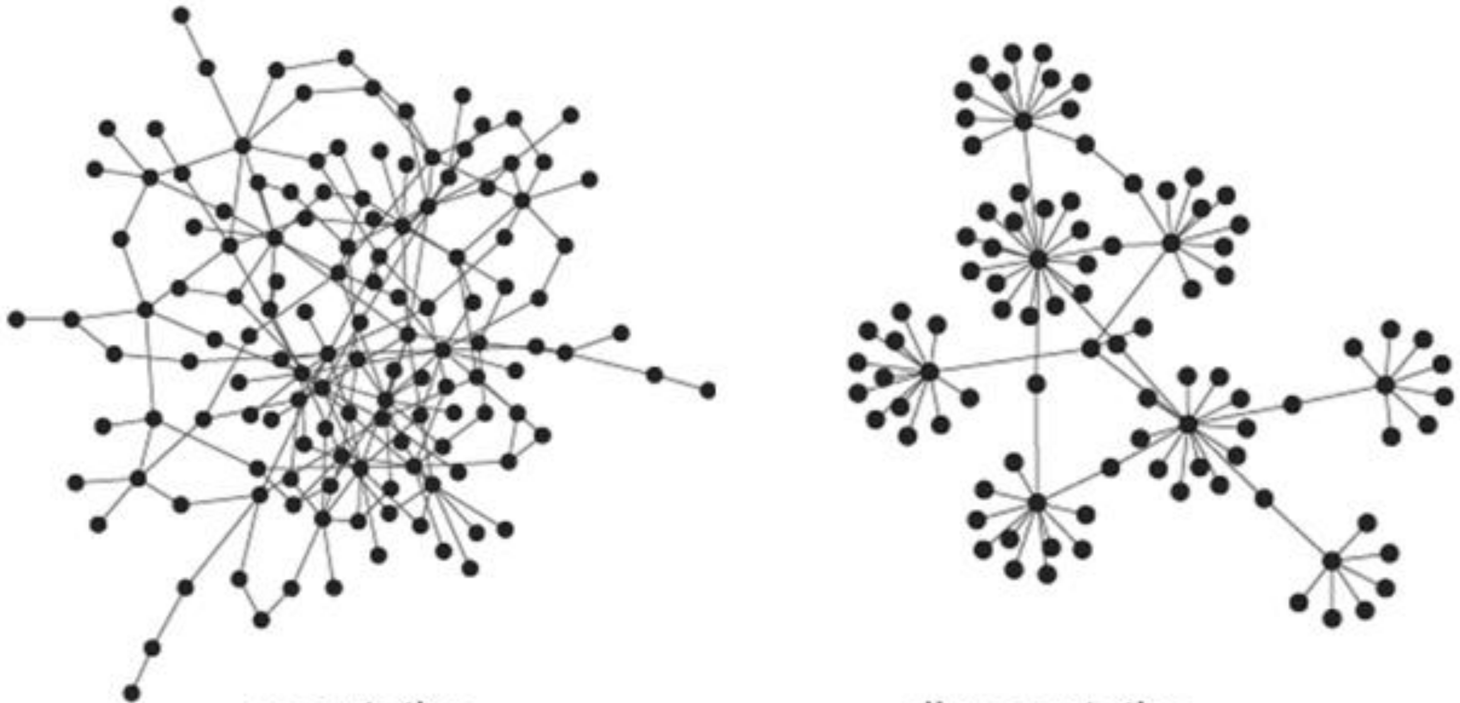
	Singtel	DBS	UOB	KrungThai
Singtel	1	0.45	0.35	0.2
DBS	0.45	1	0.6	0.3
UOB	0.35	0.6	1	0.26
KrungThai	0.2	0.3	0.26	1

Correlation Matrix - Graphs



- The width of the links shows the strengths of the correlations. Correlation matrix fundamental in portfolio risks management. Portfolio theory teaches to reduce the variance of the returns by diversification on negative correlation.
- Similarly, graphs seek to increase the overall **stability** of the whole network.

Stability of a Graphical Network



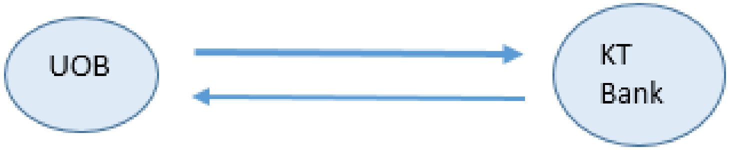
assortative disassortative

- Two types of graphs – assortative and disassortative showing the links between the portfolio assets
- Which graph network do you think will be more stable?

Something more about graphs

- Unique characteristics compared to simple correlation matrix.

Directional orientation. No such idea in correlation. Both impact each other equally. In graphs (and in practice), the impacts can be unidirectional or bi-directional.



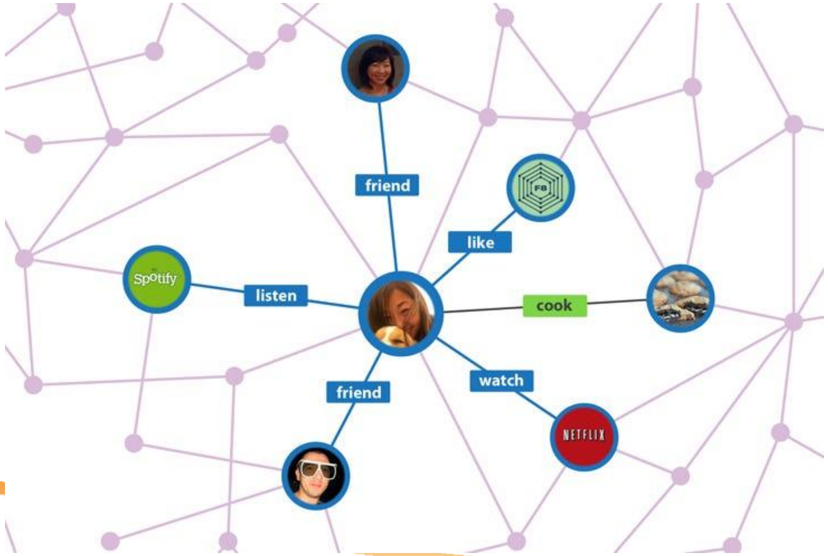
- ***Cyclical loops***
  - Either acyclical or cyclical graphs. Cyclical graphs form loops amongst the nodes resulting in feedbacks. Feedback effects more often than not in finance. Does feedback contribute to stability?
  - Eg. Does the market drive sentiment or does sentiment drive market?

Graphs in Industry

- Widely used in Technology industry. The Big Tech companies are storing data as graphs.
- Facebook social graph : data API in graphical format

Google Knowledge Graph Search API : provides for the related insights to the entities you are looking at.

- Natural fit for Big Data to ***link up*** data together from different sources and see relations



Graph Analytics – the 3 C's

Centrality

- •Which node is the important or most influential? Who are the influencers?
- •Which node impacts most on the network stability?

Clusterness

- •Which are the group of friends or social network – FB?
- •Which nodes lie outside of a cluster – outliers analysis for fraud analytics?
- •Which are the key trading hubs (clusters) in financial lending or trading with the most common trading partners?
- •Which assets in the portfolio share common risk factors or traits?
- •Where are the risk concentrations in portfolio risk management?

Graph Analytics – the 3 C's

Connectedness

- What is the influence one node has on another particular node?
- How will *news effect* cascade from a main node to another ?
- What is the shortest route from one node to another?
- What are the associated search entities with a search target entity – eg Google search recommendations
- What are the connected ones in a recommendation engine?
- In KYC, how closely related are other entities to a 'black-listed' entity?

The list of use examples can go on.
Some of these *analysis* already can be gleaned from good visuals – gephiz, d3.js, alchemy.js

Examples of Graphs AI in Finance

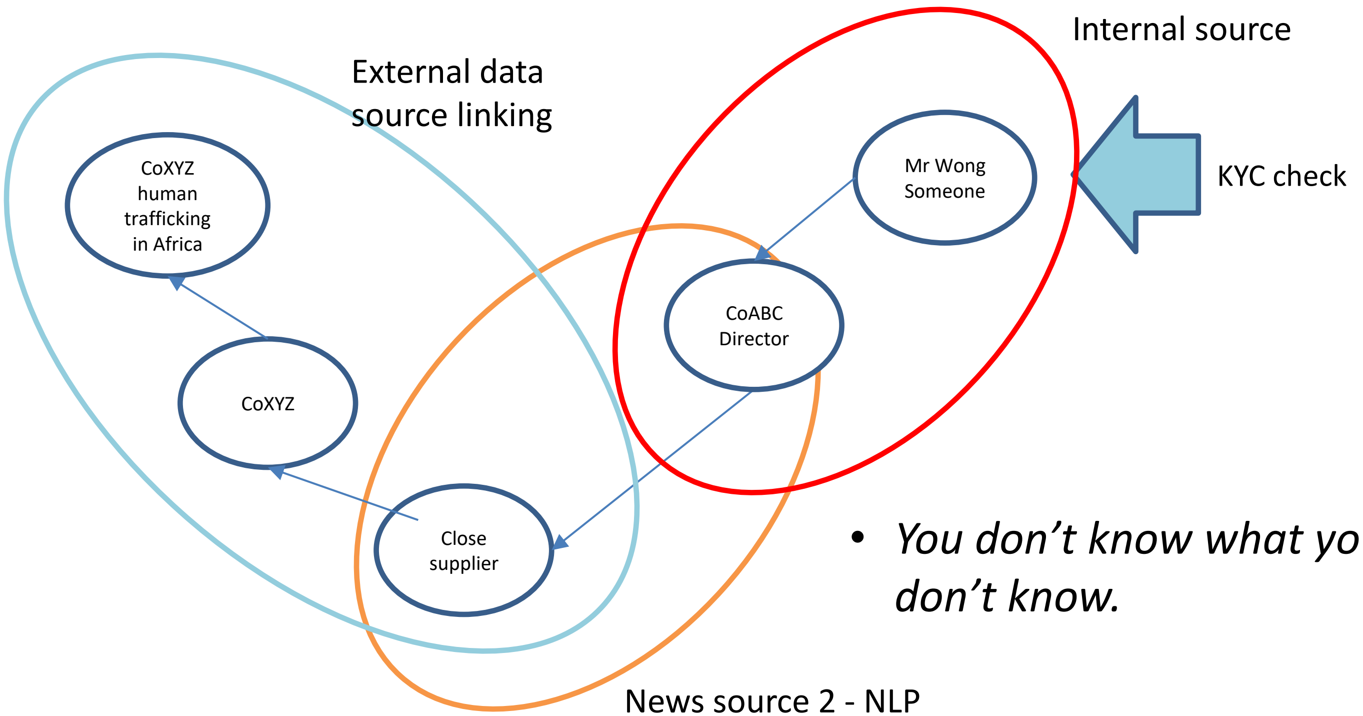
Graphical AI applied in different areas of finance.

- Financial Stability
- *BlockChain* real-time settlement (Project Ubin II)
- KYC detection
- Early warning of portfolio risks systems

… … and more

Graphs+RegTech

- Background checking more extensive using a graph of relationships
- Amalgamates internal and external sources of information in a *'data lake'.* enabling knowledge discovery.



- *You don't know what you don't know.*

Conclusion (Financial Compliance)

- **Financial Crime & Compliance**

Financial compliance covers regulatory requirements (Basel Accord, sanctions) and financial crime prevention (AML, CFT, fraud detection)

- **Regulatory Framework & Obligations**

Core obligations include KYC (Know Your Customer), Customer Due Diligence, Suspicious Activity Reporting, record keeping, and risk-based approaches
Red flags include unusual transaction patterns, geographic indicators involving high-risk jurisdictions, and customer behavior inconsistencies.

Conclusion (AI/ML)

- **AI/ML Applications in Compliance**

•AI helps sift through massive data volumes to find patterns and anomalies, reducing false positives in transaction monitoring
•Key applications: fraud detection, customer risk assessment, and automated monitoring through NLP

- **Unsupervised ML Techniques**

K-means clustering: Groups entities with similar characteristics into fixed number of clusters, measuring similarity through *distance*
DBSCAN: Density-based clustering that identifies arbitrary-shaped clusters and handles noise
Both methods enable anomaly detection without predefined rules

- **Graph Analytics in Compliance**

Graphs represent networks of nodes and relationships, capturing directional flows and feedback effects common in finance
The 3 C's: Centrality (identifying influential nodes), Clusterness (finding groups and outliers), Connectedness (tracing influence paths)